CAMPUS HACKATHON SUBMISSION

RiskPulse AI

Explainable risk intelligence

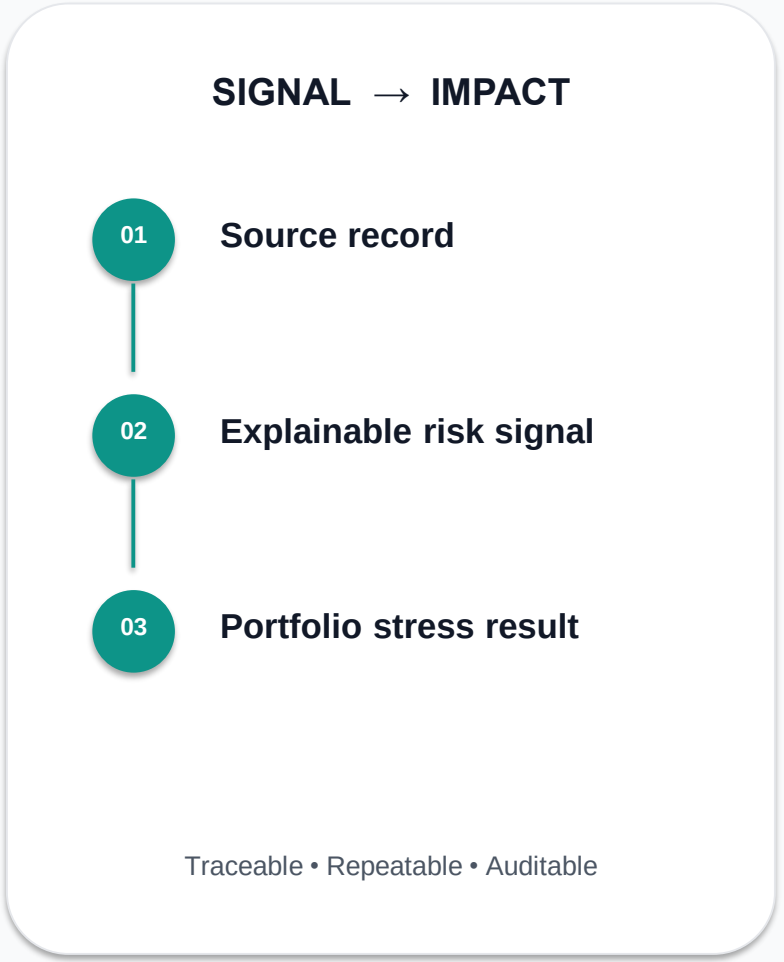
From unstructured information to auditable portfolio impact

AI / NLP risk engine + deterministic stress testing

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Turn fragmented information into actionable risk

Risk teams need a clear path from an event in the news to the exposures that may be affected.

The problem

• Information arrives as unstructured headlines and social posts.

• Entity names, event types and exposure links are easy to miss.

• Manual scenario selection can separate a stress result from its triggering evidence.

The RiskPulse AI approach

01 Ingest RSS news and historical social CSV records

02 Extract entities, sentiment and event type

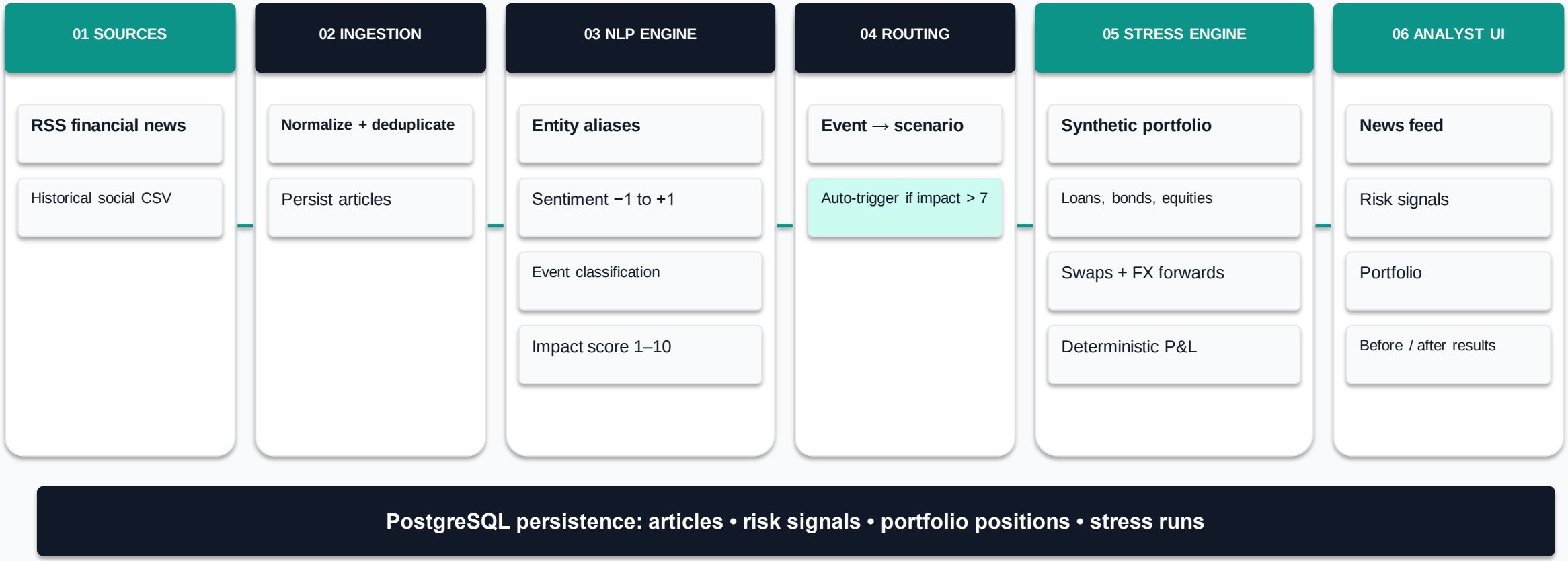
03 Score impact from 1–10 with visible factors

04 Route qualifying signals to deterministic stress scenarios

Core idea: preserve the evidence chain from source record → risk signal → affected positions → portfolio impact.

System design and data flow

Each stage produces traceable records that support the next stage.



Designed for traceability and repeatable analysis

A pragmatic stack separates ingestion, risk scoring, valuation and analyst-facing views.

Ingestion

RSS + historical social CSV

Normalization, deduplication, timestamps and bounded retries.

NLP risk engine

Entity aliases, sentiment, event classification and an impact score with factor-level explanation.

Stress engine

Event-to-scenario mapping and deterministic position-level P&L across supported asset classes.

Auditability

Persisted inputs, calculation details, triggering article links and reconciliation checks.

TECHNOLOGY

FastAPI • SQLAlchemy • PostgreSQL • Alembic • Next.js • TypeScript • Recharts

Design principle: explain what happened, why it mattered, and which positions were affected.

Current prototype: implementation evidence

Figures below are reported by the submitted project; benchmark claims are intentionally limited.

82	2	20	> 7	1.0000
Backend tests passing	Data sources in demo	Seeded portfolio positions	Default auto-trigger rule	Smoke accuracy / macro-F1*

Naive workflow

- Raw articles with no exposure linkage
- Inconsistent scores without rationale
- Manual scenario selection
- No position-level reconciliation

RiskPulse AI

- Issuer and ticker aliases link signals to positions
- Sentiment, event type and impact factors are visible
- Impact above 7 routes a scenario automatically
- Before/after P&L and matched exposures are inspectable

** PhraseBank smoke fixture: three rows only. These metrics are not a full benchmark or evidence of generalization.*

A practical analyst workflow

The system is designed to make risk triage more explainable—not to replace analyst judgement.

01 **DETECT**

A news article or historical social record enters the normalized pipeline.

02 **EXPLAIN**

Entity, event type, sentiment and impact factors are made visible.

03 **PRIORITIZE**

A configured threshold identifies material signals and routes a scenario.

04 **REVIEW**

Analysts inspect portfolio change, top losses and matched exposures.

Business value

A common evidence chain connects information, risk signals, exposures and scenario results.

Demo value

A reviewer can replay records and trace a qualifying event to the positions included in the stress result.

Prototype boundaries and next steps

RiskPulse AI demonstrates traceability and scenario mechanics; it is not a calibrated production risk model.

Current limitations

• Portfolio holdings are synthetic; transaction categories are proxies, not proof of real exposures.

• Stress shocks are illustrative assumptions, not forecasts or investment advice.

• Historical social replay does not reproduce live delivery order, deleted posts or engagement data.

• Heuristic sentiment is the safe default; FinBERT needs model setup and review.

Next steps

• Integrate permissioned portfolio and reference data.

• Calibrate shocks and validate against approved market and credit methodologies.

• Add authentication, role-based access, secrets management and observability.

• Expand model evaluation beyond the three-row smoke fixture and monitor drift.

Demo sequence: ingest records → process NLP → open Stress Testing → inspect trigger and matched positions